NAME

mbnavlab2fnv − converts a Kongsberg Navlab smoothed navigation binary file to an MB-System fast navigation (fnv) text file.

VERSION

Version 5.0

SYNOPSIS

mbnavlab2fnv [**−I***file*] [**−O***file*] [**−V**] [**−H**]
mbnavlab2fnv [**−−input**=*file*] [**−−output**=*file*] [**−−verbose**] [**−−help**]

DESCRIPTION

mbnavlab2fnv converts a Kongsberg Navlab smoothed navigation binary file (typically named *navlab_smooth.bin*) to an MB-System fast navigation (fnv) ASCII text file.

Kongsberg Navlab is a post-processing software package that computes a smoothed navigation and attitude solution for Kongsberg Hugin autonomous underwater vehicles (AUVs). The output binary file contains a time series sampled at 100 Hz (10 ms intervals) with no file header. Each record consists of 21 IEEE 754 double-precision (64-bit) floating-point values stored in little-endian byte order (record size: 168 bytes). The fields extracted by **mbnavlab2fnv** are:

field 0: time (Unix epoch seconds)
field 1: latitude (decimal degrees, positive north)
field 2: longitude (decimal degrees, positive east)
field 3: depth (meters, positive down) − used as draft
field 6: heading (degrees, 0−360, clockwise from north)
field 7: roll (degrees, positive starboard up)
field 8: pitch (degrees, positive bow up)
field 10: speed through water (m/s)

Heave is not stored in the navlab_smooth.bin format and is written as 0.0 in the fnv output.

The fnv format is a tab-separated ASCII text file. The first line is a comment beginning with **##** that names the columns. Each subsequent data line contains the following tab-separated fields:

yyyy mm dd hh mm ss.ssssss
epoch_seconds
longitude (deg)
latitude (deg)
heading (deg)
speed (km/hr)
draft (m)
roll (deg)
pitch (deg)
heave (m)

The date and time in the first field are space-delimited within their tab field. Speed is converted from m/s to km/hr. The optional port and starboard swath edge columns present in some fnv files are not written by **mbnavlab2fnv** as no swath geometry is available from the navlab_smooth.bin format.

If no input file is specified, **mbnavlab2fnv** reads from standard input. If no output file is specified, it writes to standard output. Bytes are automatically swapped on big-endian host architectures so that the output is always correct on any platform.

Records with retrograde (non-increasing) timestamps are silently skipped unless verbose mode is active.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−H

−−help
Prints the program name, usage, and a description of the navlab_smooth.bin field layout, then exits.

−V

−−verbose
Increases the verbosity of **mbnavlab2fnv**. With this option the program prints the input and output file names to standard error when they are opened, reports any skipped records with their timestamps, and prints a final summary of the number of records read and written.

−I *file*

−−input=*file*
Sets the path of the input Kongsberg Navlab binary file. If this option is not given, the program reads from standard input.

−O *file*

−−output=*file*
Sets the path of the output fnv file. If this option is not given, the program writes to standard output.

NAVLAB BINARY FORMAT

The navlab_smooth.bin format has the following properties:

- No file header of any kind.
- Each record is exactly 168 bytes: 21 values × 8 bytes (float64).
- All values are IEEE 754 double-precision floating point, little-endian byte order.
- Nominal sample rate: 100 Hz (10 ms steps).
- The complete field inventory is:

```
field  0: time_d      − Unix epoch seconds
field  1: latitude    − decimal degrees, positive north
field  2: longitude   − decimal degrees, positive east
field  3: depth       − meters, positive down
field  4: unknown     − (not used)
field  5: unknown     − (not used)
field  6: heading     − degrees, 0−360, clockwise from north
field  7: roll        − degrees, positive starboard up
field  8: pitch       − degrees, positive bow up
field  9: vert_vel    − vertical velocity, m/s, positive down
field 10: speed       − speed through water, m/s
field 11: speed_2     − speed through water, m/s (alternate source)
field 12: unknown     − (not used)
```

field 13: unknown − (not used)
field 14: std_lat − position uncertainty, latitude (deg)
field 15: std_lon − position uncertainty, longitude (deg)
field 16: std_hdg − heading uncertainty (deg)
field 17: std_17 − velocity/attitude uncertainty
field 18: std_18 − velocity/attitude uncertainty
field 19: std_19 − velocity/attitude uncertainty
field 20: cov_20 − Kalman covariance cross-term

EXAMPLES

Convert a navlab binary file to an fnv file:

```
mbnavlab2fnv −I navlab_smooth.bin −O mission.fnv
```

Use standard input and output (e.g. within a pipeline):

```
cat navlab_smooth.bin | mbnavlab2fnv > mission.fnv
```

Convert with verbose progress reporting:

```
mbnavlab2fnv −−verbose −−input=navlab_smooth.bin −−output=mission.fnv
```

Use the fnv output to supply navigation, attitude, sensor depth, and heading to **mbpreprocess** for preprocessing Kongsberg KMALL sonar data:

```
mbnavlab2fnv −I navlab_smooth.bin −O mission.fnv
mbpreprocess −−format=261 −−input=datalist.mb−1 \
    −−nav−file=mission.fnv −−nav−file−format=9 \
    −−sensordepth−file=mission.fnv −−sensordepth−file−format=9 \
    −−attitude−file=mission.fnv −−attitude−file−format=9 \
    −−heading−file=mission.fnv −−heading−file−format=9
```

SEE ALSO

mbsystem(1), **mbfnv2navlab**(1), **mbpreprocess**(1), **mbprocess**(1)

BUGS

Heave is not available in the navlab_smooth.bin format and is always written as 0.0 in the fnv output. Applications that require heave (e.g. for motion correction of multibeam bathymetry) should obtain heave from the raw sonar data stream instead.

The optional port and starboard swath edge columns in the fnv format are not written by **mbnavlab2fnv**; they are only meaningful in fnv files generated from processed swath data.